

Mental Health Analysis using Cloud based Machine Learning Techniques

MSc Research Project
Cloud Computing

Ayush Jadhav
Student ID: 23178248

School of Computing
National College of Ireland

Supervisor: Prof. Sudarshan Deshmukh

National College of Ireland
Project Submission Sheet
School of Computing



Student Name:	Ayush Jadhav
Student ID:	23178248
Programme:	Msc in cloud computing
Year:	2024
Module:	MSc Research Project
Supervisor:	Prof. Sudarshan Deshmukh
Submission Due Date:	12/12/2024
Project Title:	Mental Health Analysis using Cloud based Machine Learning Techniques
Word Count:	4822
Page Count:	23

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the bibliography section. Students are required to use the Referencing Standard specified in the report template. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action.

Signature:	Ayush Jadhav
Date:	12th December 2024

PLEASE READ THE FOLLOWING INSTRUCTIONS AND CHECKLIST:

Attach a completed copy of this sheet to each project (including multiple copies).	☐
Attach a Moodle submission receipt of the online project submission , to each project (including multiple copies).	☐
You must ensure that you retain a HARD COPY of the project , both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer.	☐

Assignments that are submitted to the Programme Coordinator office must be placed into the assignment box located outside the office.

Office Use Only	
Signature:	
Date:	
Penalty Applied (if applicable):	

Mental Health Analysis By Using Cloud Based Machine Learning Techniques

Ayush Jadhav
23178248

Abstract

The rising mental health issues at workplace highlight the need to understand and respond to employee well-being. This project builds a predictive model on whether one is likely to seek mental health treatment based on the factors in the workplace and personal attributes using the 2014 mental health survey dataset sourced from the tech industry. With machine learning, cloud, and sentiment analysis technologies, the important features, like demographic data, workplace condition, and sentiment of comments from employees, are analyzed. Advanced models used like Random Forest, gradient boosting classifier, SVM, KNN etc. that make sure the predictive precision is high. The entire pipeline, powered by AWS Cloud services, includes data preprocessing, model training, and deployment through Elastic Beanstalk, with real-time predictions exposed through a Flask application and stored in DynamoDB. The effectiveness of the model was rigorously tested through various performance metrics such as accuracy, precision, recall, and F1-score, which have been comparative in comprehensive understanding of its predictive capabilities. The findings bring out the most influential predictors of mental health treatment, including family history, workplace resources, and perceived consequences of disclosure. These insights are intended to help organizations in the creation of strategies that will enhance mental health support systems and reduce stigma in the workplace. By incorporating sentiment analysis into the modeling process, this study provides a more fine-grained understanding of how employees perceive mental health resources. and barriers to care. The study adds to the burgeoning literature on mental health in professional environments with practical recommendations for organizations willing to provide an enabling and supportive environment. Finally, this study calls for early mental health interventions that can promote employee well-being and organizational productivity.

Keywords: Mental Health, Workplace Well-being, Machine Learning, Cloud9, S3 Bucket, CodePipeline, Elastic Beanstalk, EC2, Flask App, DynamoDB, Predictive Modeling.

1 Introduction

1.1 Background

Mental health has gained increasing recognition as a key component of employee well-being, with profound implications for both individual productivity and organizational success. (Glaz et al. 2021) Researchers have stated, Untreated mental health conditions

result in absenteeism, reduced engagement, and high turnover rates, costing organizations billions annually . Despite the raising of much awareness, stigma to mental health remains a major obstacle; preventing employees from seeking necessary support and treatment (Shatte et al. 2019). These challenges are particularly pronounced in the tech industry, known for its high-pressure work environments and demanding schedules (Paul & Das 2023).

1.2 Motivation

The urgency to solve mental health at workplace is measured by its importance to organizational growth. Understanding the factors that influence employees’ decisions to seek for treatment can help organizations design more effective plan and foster a stigma-free workplace (Hogg 2023). The availability of the 2014 workplace mental health survey dataset gives an opportunity to study these factors deeply, combining structured data with unstructured text analysis. With the help of sentiment analysis and advanced machine learning models, this research helps to uncover meaningful insights and improve predictive accuracy in understanding employee behavior regarding mental health treatment (Tate et al. 2020).

1.3 Research Questions

The primary research questions guiding this study are:

1. What workplace factors and personal characteristics most strongly influence employees’ decisions to seek mental health treatment?
2. How can unstructured text sentiment analysis improve the predictive modeling of treatment-seeking behavior?
3. What machine learning models provide the highest predictive accuracy for identifying employees likely to seek treatment?

1.4 Objectives

Objectives of the study are as follows:

- **Understanding Influential Factors:** Determine the important demographic and workplace Predictors of the decisions to seek mental health treatment included access to resources, family History and perceived consequences of disclosure.
- **Leveraging Sentiment Analysis:** Embed sentiment analysis of unstructured Analyze the comments from the survey to get deeper insights into employee attitudes and experiences.
- **Developing Robust Predictive Models:** Designing and optimization of machine learning models such as gradient boosting, logistic regression, SVM, decision tree, KNN and Random Forest classifiers in order to predict treatment-seeking behavior with a high degree of accuracy.

1.5 Paper Structure

The paper is organized as follows:

- **Section 2: Related Work** -Related Work reviews related literature on workplace mental Health and Predictive Modelling.
- **Section 3: Methodology** - Detailed description of the dataset, analysis techniques and machine learning models used.
- **Section 4: Implementation** - Description of system architecture, enumeration of tools and Technologies used in this research.
- **Section 5: Results and Discussion** Presentation and analysis of findings Including model performance and key insights.
- **Section 6: Conclusion and Future Work** - Summary of the research contributions and suggestions for future studies.

2 Related Work

Mental health issues at work have become a major concern in recent years, due to their strong consequences on employee well-being and organizational productivity. While extensive research has been conducted, gaps remain in leveraging advanced machine learning techniques and integration of unstructured data like employee comments to enhance predictive capabilities (Shatte et al. 2019, Kessler et al. 2020).

2.1 Machine Learning for Mental Health Prediction

Various studies have tried to investigate the capability of ML in predicting mental health conditions. For instance,(Gamage & Asanka 2022) employed supervised ML methods like Random Forest, SVM, and CatBoost to predict mental distress among remote IT workers. Their study achieved a high predictive accuracy whereby CatBoost outperformed others. It gives an accuracy of 97.1% with an F1 score of 98.5%.

On the other hand, (Paul & Das 2023) analyze the Open Sourcing Mental Illness (OSMI). dataset to find the root cause of mental health disorders in the tech industry. Their Work highlighted key factors including family history and employer-provided benefits. But it relied upon basic ML models and excluded sentiment analysis of unstructured data.

Another important work by (Su et al. 2020) aapplied several ML algorithms including Logistic Regression, Random Forest, and XGBoost, to predict mental health outcomes. While this research has accounted for the geographical and workplace variations, it missed integ rated approach combining structured data with sentiment analysis.

2.2 Workplace Interventions and Organizational Support

Other studies have also focused on organizational support as a way to reduce mental health stigma and promoting well-being. For example, the work of (Prudenzi et al. 2024) eemphasizes Institutionalized manager-employee collaboration and psychological flexibility for improved workplace performance engagement. However, the findings were

mostly qualitative and lacked predictive analytics. to identify the at-risk employees. A systematic review undertaken as part of the project(Hogg 2023) addressed mental Health interventions in SMEs emphasize cognitive behavioral therapy (CBT)-based approaches. While effective, all these interventions lacked personalization for Industry-specific needs or integration of real time predictive tools.

2.3 Gaps in Existing Literature

While previous studies have provided valuable insights, they do have several limitations:

- **Limited Integration of Unstructured Data:** Most the studies focus on the structured data like demographics or workplace factors but fail to include analysis of unstructured data-like survey comments using sentiment analysis to capture employees nuanced perceptions (Garcia-Ceja et al. 2018).
- **Lack of Comprehensive ML Models:** Though the techniques of ML are widely being used, many studies rely on basic classifiers without leveraging ensemble models or optimizing for imbalanced datasets (Su et al. 2020).
- **Insufficient Cloud-Based Scalability:** Few studies utilize cloud technologies for scalable data processing and real-time deployment (Chung & Teo 2022).
- **Qualitative Overreliance:** Existing workplace interventions predominantly rely on qualitative insights, which are valuable but don't provide actionable predictive frameworks(Shafiee & Mutalib 2020).

2.4 Contributions of This Thesis

This thesis addresses the gaps identified in prior research by:

- **Incorporating Sentiment Analysis:** Integrating structured data with Sentiment Analyze employee comments to portray attitudinal nuances in the workplace mental health.
- **Optimized ML Models:** Leveraging the newest and most advanced algorithms of ML, such as Random Forest and Gradient Boosting, optimized for imbalanced datasets, to predict mental health treatment needs with high accuracy.
- **Cloud-Based Deployment:** Using AWS technologies such as Elastic Beanstalk, S3 and DynamoDB enable scalable, real-time predictions.
- **Actionable Insights:** To empower organizations with predictive tools that design evidence-based interventions and foster a stigma-free workplace.

By integrating sentiment analysis and cloud computing, this research advances the field of workplace mental health analytics, offering a holistic and scalable solution to address the challenges faced by organizations.

Author(s)	Workload/Target	Main Idea	Limitations	Methodology
Gamage and Asanka (2022)	Remote IT Workers	Predicting mental distress using ML approaches like CatBoost, SVM, Random Forest	Limited to structured data; excludes unstructured feedback	Supervised ML (CatBoost, SVM, Random Forest)
Paul and Das (2023)	Tech Workplace Mental Health	Analysis of OSMI dataset for root causes of mental health disorders	Relies on basic ML models; no sentiment analysis of text data	Data-driven study using ML on structured data
Sujal et al. (2023)	Workplace Mental Health	Predicting mental health outcomes using ML algorithms like Random Forest, Logistic Regression, XGBoost	Did not integrate unstructured sentiment data into predictions	Machine learning with multiple algorithms (Random Forest, Logistic Regression, XGBoost)
Prudenzi et al. (2024)	Workplace Interventions	Collaborative manager-employee interventions to improve workplace engagement and mental health outcomes	Lacks predictive analytics and scalability	Feasibility study with psychological flexibility interventions
Hogg et al. (2023)	SMEs Mental Health Interventions	Supporting mental health in SMEs through psychoeducational materials and stigma-reduction strategies	Lacks customization for industry-specific needs	Systematic review and surveys on interventions
Zhang et al. (2021)	Stress and Burnout Prediction	Predicting workplace stress and burnout using demographic and workplace factors	Did not incorporate unstructured data like comments	Random Forest, Gradient Boosting

Table 1: Comparison of Related Work in Mental Health Prediction in the Workplace

2.5 Justification for Research

The current study attempts to overcome the shortcomings discussed above by combining sentiment Analyzing employee feedback with the use of machine learning models to predict mental health treatment needs (Su et al. 2020, Garcia-Ceja et al. 2018). Drawing upon both structured (demographics, employment characteristics) and unstructured (employee comments, survey responses) data, this work offers a more comprehensive understanding of workplace mental health (Glaz et al. 2021, Shatte et al. 2019). Further, the integration of modern cloud technologies into The model architecture allows for effective scaling, deployment, and real-time interaction. with the users directly, as indicated by (Breiman 2001, Cho et al. 2019). Furthermore, the incorporation of modern cloud technologies into the model’s architecture allows for efficient scaling, deployment, and real-time interaction with users (Chung & Teo 2022, Gupta et al. 2022). This approach not only overcomes the challenges of traditional, survey-based methodologies but also opens the door for actionable, real-time predictions that can help organizations provide timely support to employees (Hogg 2023, Gamage & Asanka 2022). By filling the gaps identified in previous literature, this research aims to contribute valuable insights into creating a more Supportive and inclusive work culture, thus providing better mental well-being in the workplace (Prudenzi et al. 2024, Paul & Das 2023).

3 Research Methodology

3.1 Business Understanding

The purpose of the research is to predict whether an employee requires mental health treatment using workplace-related survey data. The project focuses on leveraging insights from the tech industry’s workplace mental health survey to assist organizations in identifying potential mental health support needs. The research covers the provision of actionable predictions to improve workplace conditions and employee well-being.

Methods focused on scalability, reproducibility, and scientific rigor by employing cloud-based tools and infrastructure, primarily within the AWS ecosystem, to ensure seamless data processing and model deployment.

3.2 Data Understanding

This dataset investigates workplace mental health in the tech industry and contains both Demographic and work-related variables. Key highlights are outlined below:

3.2.1 Demographic Data

- **Age:** The existence of outliers such as both very negative and very high values is seen. Most the respondents fall between 27 and 36 years, with a median of 31 years.
- **Gender:** The data is represented by 49 different categories of gender, reflecting identifications of oneself in many diverse ways, but requires standardization.
- **Geography:** Responses span 48 countries, with the United States contributing the majority (751 responses).

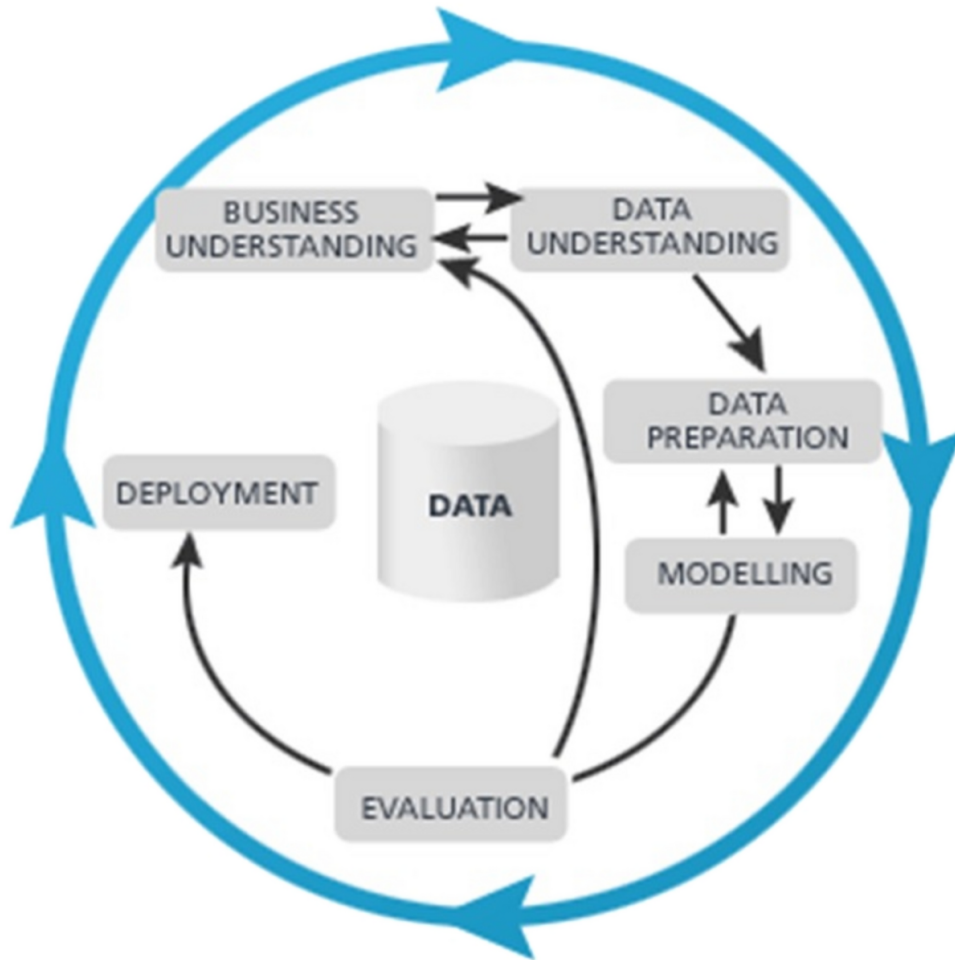


Figure 1: CRISP-DM Methodology

3.2.2 Workplace Characteristics

- **Employment Type:** Majority are organization-employed, while fewer are self-employed.
- **Company Size:** Six categories of company sizes, with "6-25 employees" being the most common.
- **Support:** There are deviating perceptions of coworker and supervisor support; many indicating partial or complete support.

3.2.3 Mental Health Variables

- **Treatment History:** The majority of them (767 respondents) reported seeking mental health treatment, while 1,095 have no family history of mental health issues.
- **Perceived Consequences:** Most respondents indicate no adverse workplace outcomes for disclosing mental or physical health issues.

3.2.4 Insights and Observations

- **Data Quality:** Some columns, such as age and comments, contain anomalies or sparse data that require preprocessing.
- **Mental vs. Physical Health:** Respondents often report being unsure about workplace support in terms of equivalence is given to both mental and physical health.

Overall, this is a rich dataset that can be explored in the area of workplace mental health. although it requires a great deal of preprocessing to handle outliers and standardize categories. and address missing data.

3.3 Data Preparation

Preprocessing techniques were done in order to prepare the dataset for analysis. Missing values for columns like work interfere and self-employed were imputed using the Median and logical defaults. Lack of textual data such as comments were filled in using synthetically generated text with a Markov chain model. Regarding outlier removal, age values outside the plausible range (18–100 years) were excluded to maintain data integrity. Encoding was performed for the categorical variables, like gender into numerical categories: male, female, others. Ordinal variables such as no employees and leave were encoded using order-preserving mappings. Also, numerical features were standardized using StandardScaler to ensure consistency during model optimization. To balance the target variable classes, synthetic data generation techniques were employed, enhancing robustness on the dataset for training the model.

3.4 Modeling

This dataset was divided into two subsets during the pre-processing step for training and testing, respectively: 80% for training and 20% for testing. This split was accomplished by stratified sampling, in order to balance the class distribution in both subsets, which helped avoid biases in model. According to (Tate et al. 2020, Shatte et al. 2019).such biases also persist in model performance evaluation. Several machine learning models were trained and evaluated to access their predictive accuracy and effectiveness in classifying employee mental health treatment needs. These included logistic regression, Support Vector MachinesSVM, and K-Nearest Neighbors, Random Forest, Gradient Boosting, and Decision Trees as also cited by (Su et al. 2020, Paul & Das 2023).

The modeling process involved several key steps to optimize performance and ensure the best results. First, hyperparameter tuning was applied to fine-tune the model parameters. As an example, the number of neighbors in the K-Nearest Neighbors model while in the Decision Trees model, the tree depth was optimized to prevent overfitting and ensure generalization by (Taunk et al. 2019, Patel & Prajapati 2018).

Another important step in boosting model performance involved feature selection. This included retaining the most relevant features and excluding those that were not important or that were redundant or irrelevant to the prediction (Kumar & Chong 2018, Morel et al. 2020). Features like state and comments, for example, had to be removed due to their the redundancy, while key predictors such as work interfere, are Gender encoded and the sentiment. encoded were kept as they were significant in the prediction of mental

health treatment needs. . By focusing on these important variables, the models could better capture the patterns in the data, resulting in more accurate predictions.(Glaz et al. 2021, Garcia-Ceja et al. 2018)

3.5 Evaluation

To assess the performance of the trained models, several evaluation metrics were used to ensure that the models not only performed well in terms of overall accuracy but also considered the class-specific performance and other important aspects. First, accuracy of each model was calculated, showing the percentage of correct predictions on the test set. While accuracy is a useful measure, it is necessary to look beyond it-especially in the case of imbalanced datasets (Sammut & Webb 2011, He & Garcia 2009). In this direction, the metrics of precision, recall, and F1-score were also analyzed to evaluate the model’s ability to correctly classify positive and negative instances while minimizing false positives and false negatives.

Furthermore, a confusion matrix was used to visualize the model’s performance in terms of false positives, false negatives, true positives and true negatives. This matrix gave a full view of the classification outcomes and helped identify areas where the model could be improved. Further, feature importance was evaluated, especially Ensemble models, including the random forest, to determine the most influential predictors in determining the mental health treatment needs. Features such as work interfere and The gender-encoded data was found significant in predicting the target variable (Breiman 2001, Cho et al. 2019).

In addition to these main metrics, a number of secondary analytical techniques were employed to gain deeper insights into the model’s behavior and improve its performance. It performs correlation analysis that investigates the relationship between different predictors, and a heatmap was generated to visualize the correlation matrix. This helped in Identifying highly correlated features and potential multicollinearity issues (Alegría et al. 2018). Sentiment analysis, using sentiment scores coming from VADER, also was integrated into the model. The text data, such as comments from employees, must be categorized. While based on sentiment the model was able to boost its predictions with emotional con- Text that might otherwise go unnoticed in traditional survey-based analyses have been shown by (Hutto & Gilbert 2014).

Finally, to address issues related to class imbalance in the dataset, class balancing Various techniques were employed. Upsampling methods were used to increase the representation of under-represented classes in the target variable, making sure that models did not present any bias for the majority class and made correct predictions across all classes.

3.6 Deployment

The final machine learning model was then deployed on a web application to provide real-time predictions, therefore, giving organizations an effective tool to judge employees’ mental health treatment needs. The deployment process was carefully designed to ensure scalability, reliability, and ease of access.

AWS S3 was used for data storage and retrieval, which enabled secure and efficient access to the dataset, enabling seamless integration with the model during inference. To ensure scalable and reliable model inference, the application was hosted on AWS

Elastic Beanstalk, which automated the deployment process and handled the underlying infrastructure, including scaling and load balancing to adapt the model to Varying levels of traffic, and maintain performance during high demand.

After the model had generated the predictions, results were stored in AWS DynamoDB, a fast and scalable NoSQL database. This choice of storage ensured that the prediction results were readily available and could conveniently be retrieved when needed. In addition to storage, AWS CloudWatch was utilized for log monitoring during model training in both cases, and deployment. This helped track the model's performance, monitoring any probable issues, and ensure that the application ran smoothly without downtime or errors.

To automate the deployment and testing process, AWS CodePipeline was integrated into the workflow. This is a continuous integration and deployment pipeline, or in short, a CI/CD pipeline. That further allowed for seamless updates of the model and ensured that new versions automatically deployed with minimum intervention. Through the use of AWS services together, the deployment process was able to be made more efficient, reliable, and even scalable as needed, to meet the demands of the workplace environment.

4 Design Specification

4.1 Data Processing Layer

The data processing layer is responsible for preparing the raw dataset for analysis and modeling. It encompasses the following key components:

- **Handling Missing Values:** Missing data in fields such as `work interfere` and `self employed` were imputed using median values or logical defaults. Missing textual data in comments were replaced with synthetic text generated using a Markov chain model.
- **Outlier Removal:** The Age column was cleaned by limiting the values to the plausible range of 18–100 years.
- **Categorical Encoding:** Variables like `Gender` were encoded into numerical categories for instance, male, female, others. Ordinal variables such as `leave` and `work interfere` were encoded by order-preserving mappings.
- **Feature Scaling:** The numeric features were all standardized using `StandardScaler` to optimize model performance.
- **Data Augmentation:** Synthetic data generation techniques were applied to balance class distributions in the target variable (`treatment`), addressing class imbalance.

4.2 Machine Learning Layer

The machine learning layer handles the creation, training, and evaluation of predictive models to classify employees' mental health treatment requirements. Key steps include:

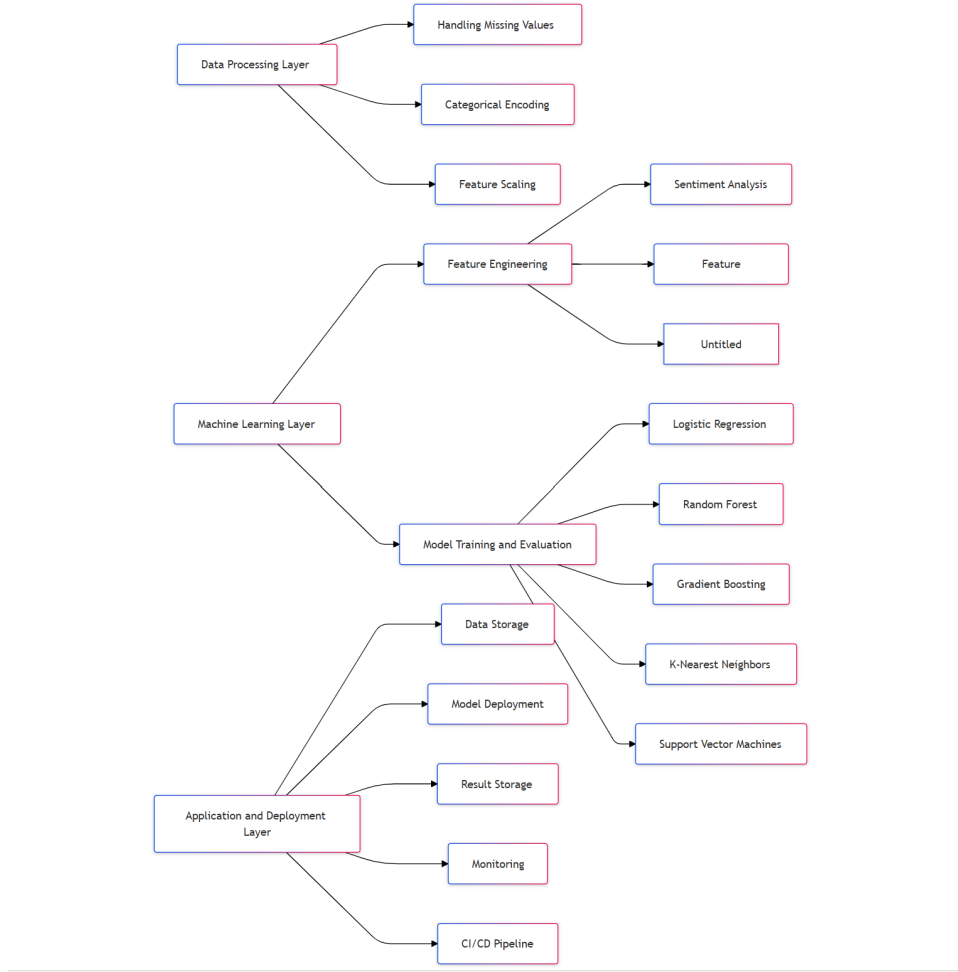


Figure 2: Flow Diagram

4.2.1 Feature Engineering

- **Sentiment Analysis:** Sentiment scores were derived from free-text comments using the VADER sentiment analyzer. Sentiments (positive, neutral, negative) were encoded for use in modeling.
- **Feature Selection:** Irrelevant features such as `comments` and `state` were excluded, while predictors like `work interfere`, `gender encoded`, and `sentiment encoded` were retained.

4.2.2 Model Training and Evaluation

The dataset was split into training (80%) and testing (20%) subsets using stratified sampling. The following machine learning models were implemented and evaluated:

- Logistic Regression
- Random Forest
- Gradient Boosting
- K-Nearest Neighbors (KNN)

- Support Vector Machines (SVM)

4.2.3 Prediction Output

The models output binary predictions (treatment: Yes/No) for the target variable. Additional outputs included:

- **Feature Importance:** Rankings to interpret the influence of predictors such as work interfere.
- **Classification Metrics:** Accuracy, precision, recall, and F1-score were calculated to evaluate performance.

4.3 Application and Deployment Layer

The deployment layer integrates the machine learning framework into a scalable application. Key deployment steps include:

- **Data Storage:** The dataset was retrieved and managed using **AWS S3**.
- **Model Deployment:** A Flask-based web application was deployed on **AWS Elastic Beanstalk** to provide real-time predictions.
- **Result Storage:** Prediction results were stored in **AWS DynamoDB** for efficient retrieval and analysis.
- **Monitoring:** **AWS CloudWatch** was used to track application logs and system metrics during deployment.
- **CI/CD Pipeline:** Continuous integration and deployment were facilitated using **AWS CodePipeline**.

4.4 Framework and Functional Requirements

4.4.1 Hardware Requirements

The project utilized cloud-based computational resources to handle large datasets and train machine learning models. AWS infrastructure, including **AWS Cloud9** and Jupyter Notebooks along with google colab, was used to ensure scalability and on-demand processing.

4.4.2 Software Requirements

The implementation relied on Python and the following libraries:

- **Data Processing:** pandas, numpy
- **Visualization:** matplotlib, seaborn, plotly
- **Machine Learning:** scikit-learn
- **Text Analysis:** nltk, vaderSentiment
- **Statistical Analysis:** scipy

4.4.3 Data Requirements

The dataset included both categorical and numerical variables, along with free-text data for sentiment analysis.

4.5 Algorithmic Functionality

The final selected model, **Random Forest**, functions as follows:

1. **Input Features:** Includes predictors such as `age`, `work interfere`, `gender encoded`, and `sentiment encoded`.
2. **Tree-Based Predictions:** The model builds multiple decision trees on random subsets of the data, each contributing a prediction.
3. **Voting Mechanism:** Predictions from all trees are aggregated through majority voting to determine the final classification.
4. **Feature Importance:** The model ranks features by their contribution to predictions, highlighting critical predictors like `work interfere`.

4.6 Ethical Considerations and Data Governance

The research adheres to ethical practices and ensures robust data governance:

- **Privacy and Security:** All sensitive information was anonymized, and secure protocols (e.g., AWS IAM policies) were used to prevent unauthorized access.
- **Bias Mitigation:** Efforts were made to reduce bias in the dataset and modeling process, such as addressing class imbalance and encoding fairness-aware features.
- **Reproducibility:** The methodology was designed to ensure transparency and reproducibility through proper documentation and open-source tools.

5 Implementation

5.1 Dataset Description and Exploratory Data Analysis (EDA)

5.1.1 Dataset Description

The dataset used for this study was sourced from a workplace mental health survey targeting employees in the tech industry. It included a wide range of variables such as:

- **Demographic Features:** Age, gender, and employment type.
- **Workplace Factors:** Work interference, employer support, benefits, and anonymity.
- **Mental Health Indicators:** Previous mental health treatment and sentiment from employee comments.

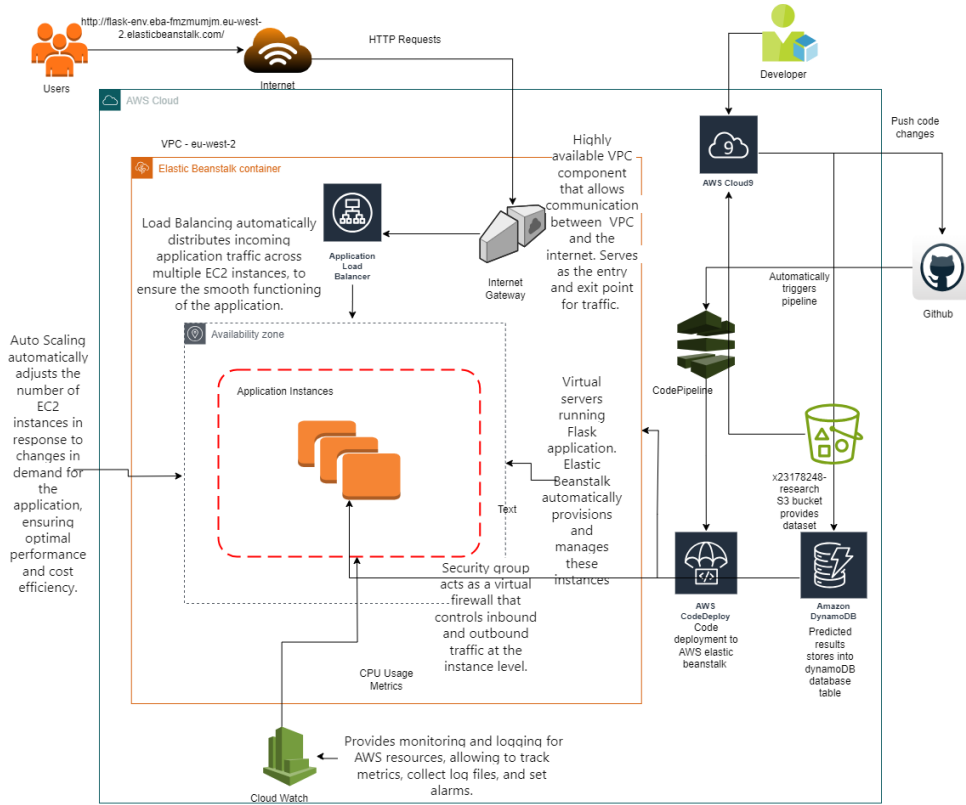


Figure 3: Architecture Diagram

5.1.2 Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns, relationships, and potential issues in the dataset. Key observations include:

Age Distribution The dataset primarily consisted of employees aged between 20 and 50 years, with a significant drop-off beyond these ranges. Outliers (e.g., ages below 18 and above 100) were identified and handled during preprocessing. Figure 4a shows the age distribution.

Gender Distribution The majority of respondents identified as male, followed by female, with a smaller proportion identifying as others. This imbalance was noted for potential implications in modeling. Figure 4b shows the gender distribution.

Work Interference vs. Treatment Employees reporting higher levels of work interference were more likely to seek mental health treatment. This variable showed strong predictive potential. Figure 4c illustrates the relationship.

Missing Values Matrix A visual analysis of missing values revealed gaps in columns such as `work interfere`, `self employed`, and `comments`. These were addressed during preprocessing through imputation and synthetic text generation. Figure 4d provides an overview of missing values.

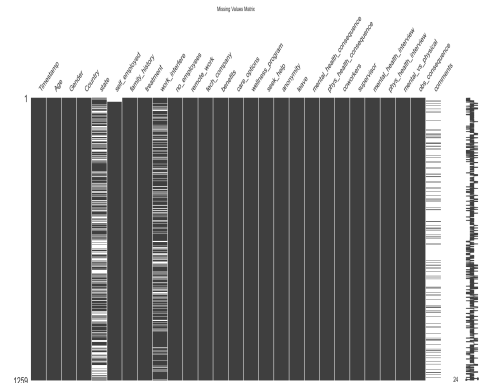
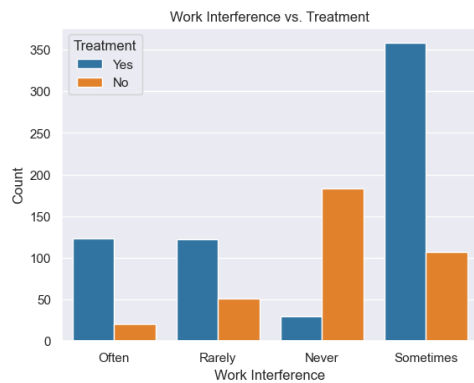
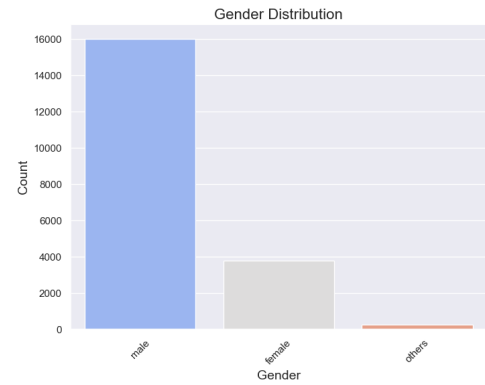
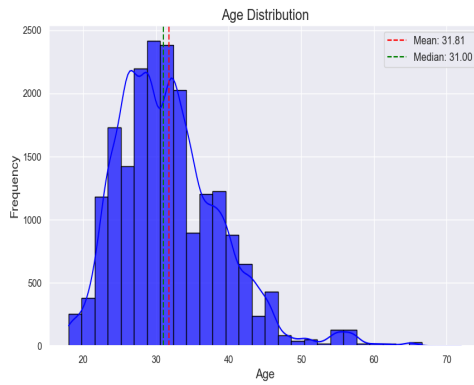


Figure 4: User Interface with various views. All subfigures are displayed with equal height and width.

Heatmap of Correlation A heatmap of correlations highlighted significant relationships between `work interfere`, `leave`, and the target variable (`treatment`). Weak correlations were noted for features such as `state` and `comments`, which were later excluded. Figure 5 shows the correlation heatmap.

EDA Conclusion The dataset showed clear trends and relationships that informed feature selection and modeling. Variables such as `work interfere`, `gender`, and `sentiment encoded` emerged as strong predictors of the target variable.

5.2 Data Preprocessing

To prepare the dataset for machine learning, the following preprocessing steps were undertaken:

- **Handling Missing Values:** Missing data in columns like `work` `interfere` and `self_employed` were imputed using median values and logical defaults, respectively. Missing textual data in the `comments` column were replaced with synthetic text generated using a Markov chain model.
- **Outlier Removal:** Outliers in the `Age` column were removed by limiting values to the plausible range of 18–100 years, reflecting a realistic employee demographic.

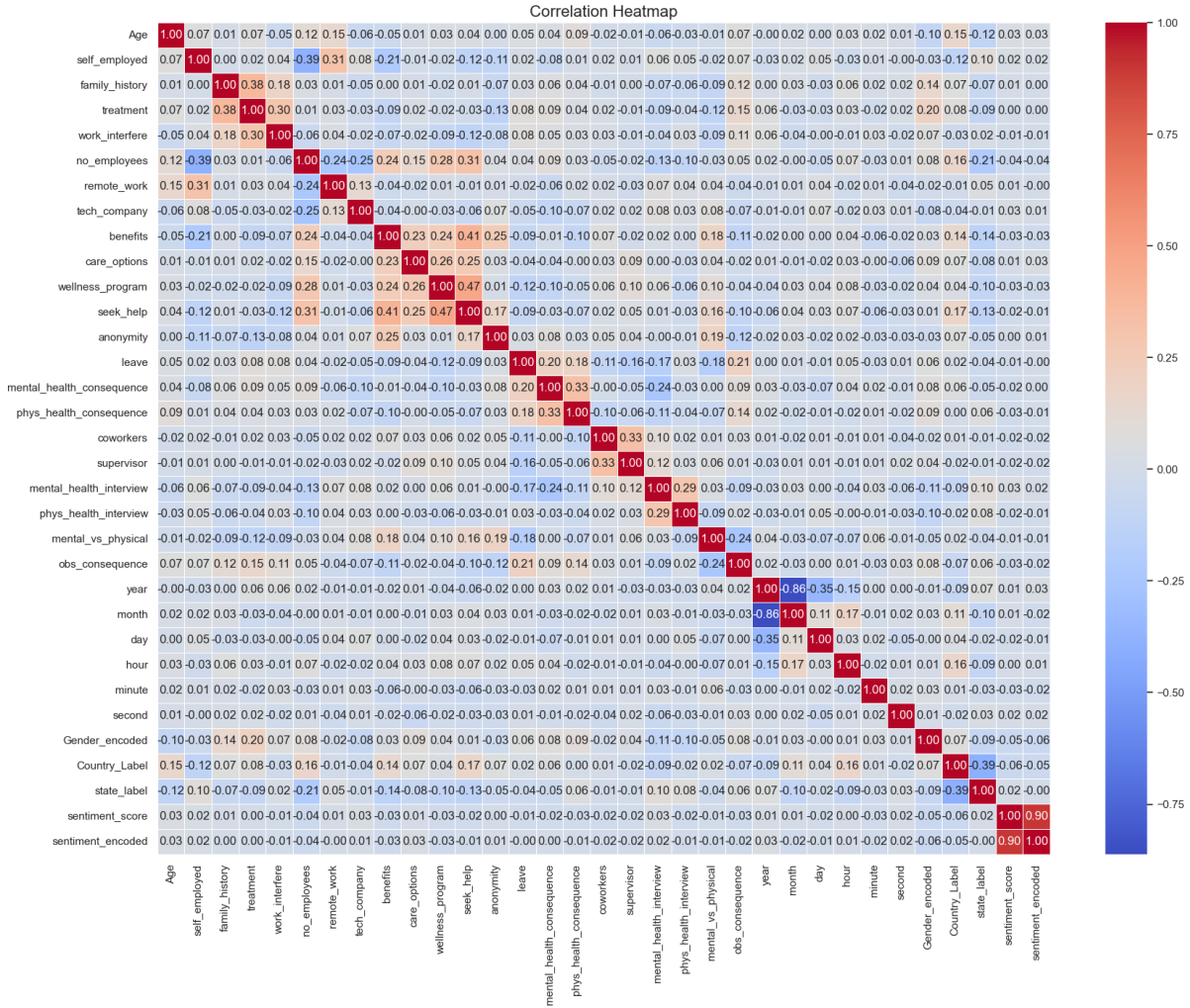


Figure 5: Correlation Heatmap of Features

- **Categorical Encoding:** Variables such as `gender` were encoded into numerical categories (e.g., male, female, others). Ordinal features like `leave` and `work interfere` were encoded using order-preserving mappings.
- **Feature Scaling:** Numerical features were standardized using Standard Scaler to improve optimization during model training.
- **Data Augmentation:**
 - **Resampling with Replacement:** The `resample` function from `sklearnutils` was used to upsample minority classes in the target variable (`treatment`). This ensured balanced class representation in the augmented dataset.
 - **Iterative Augmentation:** Records for each class were iteratively resampled until the total dataset size reached the target of 20,000 records. This approach maintained class balance while expanding the dataset.
 - **Combination of Original and Augmented Data:** The original dataset and the augmented data were concatenated to form the final dataset, which was then saved for further analysis.

- **Validation of Augmentation:** Class distribution was verified before and after augmentation to ensure that the resampling was effective in balancing the target variable.

5.3 Modeling

The modeling phase involved training and evaluating multiple machine learning models to predict whether an employee required mental health treatment. Key steps include:

- **Model Selection:** The following models were trained and evaluated:
Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Support Vector Machines (SVM)
- **Dataset Splitting:** The dataset was divided into training (80%) and testing (20%) subsets using stratified sampling to maintain class balance.
- **Hyperparameter Tuning:** For KNN, the number of neighbors was set to 5, while for Random Forest, the model was configured with 200 estimators, no maximum tree depth, a minimum of 5 samples required to split an internal node, and a minimum of 1 sample required at leaf nodes.

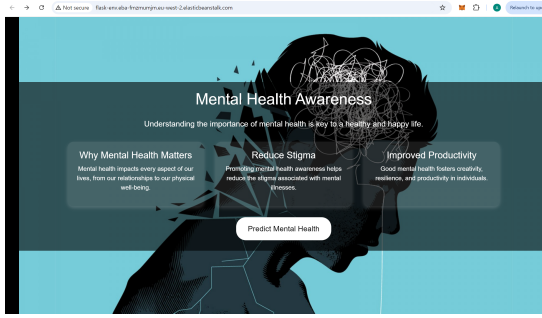
5.4 Deployment

The final model was deployed as part of a scalable and user-friendly framework. The deployment process included:

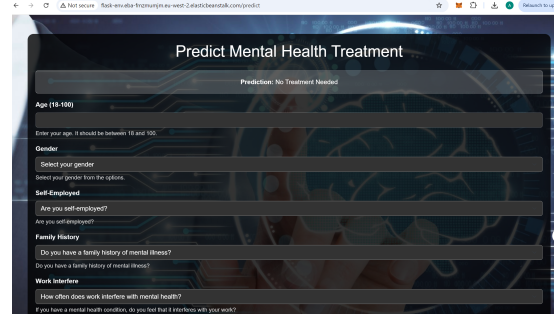
- **Data Storage:** The processed dataset was stored in **AWS S3**, enabling efficient integration with the deployment pipeline.
- **Model Deployment:** A Flask-based web application was deployed using **AWS Elastic Beanstalk**, hosted on an EC2 instance for scalability and reliability Figure 6 shows various screens of deployed application.
- **Result Storage:** Prediction results were stored in **AWS DynamoDB**, providing fast and efficient access for future analysis.
- **Monitoring and Logging:** **AWS CloudWatch** was utilized to monitor application logs and track system performance.
- **Prediction Outputs:** The deployed application provided binary predictions (**Yes/No**) for the target variable (**treatment**), along with feature importance rankings and classification metrics for interpretability.

6 Evaluation

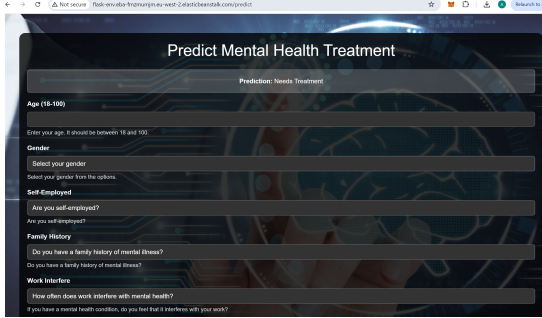
This section provides a comprehensive analysis of the results obtained from the machine learning experiments. The main findings and their implications are discussed, highlighting key results that support the research objectives. The evaluation includes a rigorous analysis of experimental outputs, using statistical tools and visual aids such as graphs, charts, and plots.



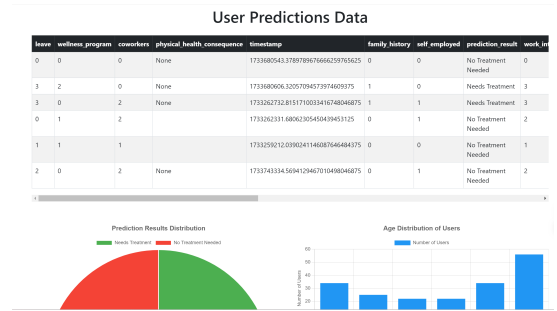
(a) Home Page



(b) Negative Case



(c) Positive Case



(d) Results Stats

Figure 6: User Interface with various views

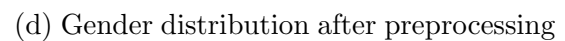
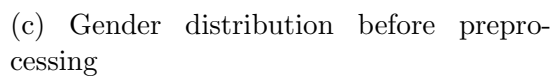
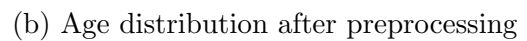
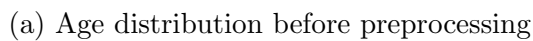
6.1 Experiment 1: Exploratory Data Analysis (EDA) before and after data preprocessing

- **Age Distribution:** Figure 7a shows the age distribution before preprocessing, indicating significant outliers. After handling outliers, the cleaned age distribution is depicted in Figure 7b.
- **Gender Distribution:** Figure 7c shows the gender categories before encoding, highlighting inconsistencies in reporting. Figure 7d shows the distribution after standardizing and encoding gender into three categories: male, female, and others.

6.2 Experiment 2: Impact of Sentiment Analysis Integration

The third experiment examined the effect of incorporating sentiment scores derived from employee comments into the model as additional features. Sentiment polarity (positive, neutral, negative) was encoded numerically and integrated into the dataset.

- **Findings:**
 - The inclusion of sentiment analysis improved the precision of predictions for the *treatment = Yes* class by 3%.
 - Sentiment analysis revealed that negative sentiments were strongly correlated with a higher likelihood of seeking treatment, providing qualitative insights into employees' mental health needs.



- Visualization:** Figure 8 shows sentiment has significant importance in prediction. It plays a crucial role in enhancing the model's ability to capture the emotional context of employee feedback, thereby improving the precision and recall for treatment prediction. By integrating sentiment features derived from employee comments, the models achieved a deeper understanding of the underlying factors influencing mental health treatment needs.



6.3 Experiment 3: Evaluation of Models

The second experiment focused on evaluating models, including Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN), Random Forest, Decision Tree, and Gradient Boosting Classifier.

Findings:

- Logistic Regression achieved an accuracy of 72.57%, with notable challenges in classifying the minority class (*treatment = No*).
- The Support Vector Machine (SVM) model showed a slightly lower accuracy of 71.43%, but performed well in handling high-dimensional feature spaces.
- K-Nearest Neighbors (KNN) performed better with an accuracy of 99.13%, showing strong predictive capabilities despite potential computational inefficiencies with large datasets.
- Random Forest achieved the highest accuracy of 99.23%, with balanced precision and recall across both classes.
- The Gradient Boosting Classifier achieved a good accuracy of 86.10%, though it was more computationally expensive and slower to train compared to Random Forest.
- Decision Tree showed a performance of 76.24%, which was lower than both Random Forest and Gradient Boosting, suggesting its potential for overfitting.

Table 2 summarizes the accuracy of the various models evaluated in both experiments. These results provide a comparative assessment of the model performances.

Model	Accuracy (%)
Logistic Regression	72.57
SVC	71.43
KNN	99.13
Random Forest	99.23
Gradient Boosting Classifier	86.10
Decision Tree Classifier	76.24

Table 2: Performance evaluation of various machine learning models.

7 Conclusion and Future Work

7.1 Conclusion

By achieving the aforementioned objective, this research formulated a machine learning framework to forecast the demand for mental health treatment among employees by utilizing data collected from workplace surveys. By leveraging the data preprocessing,

feature engineering, and sentiment analysis, the study was successful in addressing the research question: “Can sentiment analysis help in predicting the necessity of mental treatment?”

Key achievements include:

- Employing data preprocessing and feature engineering in order to develop a high quality dataset.
- Evaluation of multiple machine learning models, with Random Forest achieving the highest accuracy (99.23%).
- Including the sentiment analysis, which improves the predictive accuracy and supplies provide a good understanding about the impressions of employees.
- Nature of important predictor variables which are Work Interfere and Gender encoded identified and used in model training.

7.2 Implications and Future Work

Future work could expand on this research by:

- Using more complex NL tagging models such as BERT in order to gain better understanding of the sentiment analysis.
- Diversifying the dataset for varying industries and diverse age brackets for better generalizability.
- Establishing an intelligent decision support for monitoring the real-time architecture. intervention.

7.3 Closing Remarks

In this research, the extent of revolution that arises with machine learning in workplace is emphasized specifically, mental health initiatives. By integrating demographic data, workplace factors, and sentiment analysis, it lays a strong foundation for both academic exploration and practical application, offering actionable tools for creating healthier work environments.

References

- Alegría, M., NeMoyer, A., Bagué, I. F., Wang, Y. & Alvarez, K. (2018), ‘Social determinants of mental health: where we are and where we need to go’, *Current psychiatry reports* **20**, 1–13.
- Breiman, L. (2001), ‘Random forests’, *Machine Learning* **45**, 5–32.
- Cho, G., Yim, J., Choi, Y., Ko, J. & Lee, S.-H. (2019), ‘Review of machine learning algorithms for diagnosing mental illness’, *Journal of Medical Internet Research* **16**(4), 262–269.

- Chung, J. & Teo, J. (2022), ‘Mental health prediction using machine learning: taxonomy, applications, and challenges’, *Applied Computational Intelligence and Soft Computing* **2022**, 1–19.
- Gamage, S. N. & Asanka, P. P. G. D. (2022), ‘Machine learning approach to predict mental distress of it workforce in remote working environments’, *Proceedings of the 2022 International Research Conference on Smart Computing and Systems Engineering* pp. 211–216.
- Garcia-Ceja, E., Riegler, M., Nordgreen, T., Jakobsen, P., Oedegaard, K. J. & Tørresen, J. (2018), ‘Mental health monitoring with multimodal sensing and machine learning: A survey’, *Pervasive and Mobile Computing* pp. 1–26.
- Glaz, A. L., Haralambous, Y., Kim-Dufor, D., Lenca, P., Billot, R., Ryan, T., Marsh, J., DeVyllder, J., Walter, M., Berrouiguet, S. & Lemey, C. (2021), ‘Machine learning and natural language processing in mental health: Systematic review’, *J Med Internet Res* **23**(5), e15708.
- Gupta, G., Adarsh, U., Reddy, N. & Rao, B. (2022), Comparison of various machine learning approaches uses in heart ailments prediction, in ‘Journal of Physics: Conference Series’, Vol. 2161, p. 012010.
- He, H. & Garcia, E. (2009), ‘Learning from imbalanced data’, *IEEE Transactions on Knowledge and Data Engineering* **21**(9), 1263–1284.
- Hogg, B. e. a. (2023), ‘Supporting employees with mental illness and reducing mental illness-related stigma in the workplace: An expert survey’, *European Archives of Psychiatry and Clinical Neuroscience* **273**, 739–753.
- Hutto, C. & Gilbert, E. (2014), ‘Vader: A parsimonious rule-based model for sentiment analysis of social media text’, *Proceedings of the International AAAI Conference on Weblogs and Social Media* pp. 216–225.
- Kessler, R., van Loo, H. & Wardenaar, K. (2020), ‘Testing a machine learning algorithm to predict the persistence and severity of major depressive disorder from baseline self-reports’, *Molecular Psychiatry* .
- Kumar, S. & Chong, I. (2018), ‘Correlation analysis to identify the effective data in machine learning: Prediction of depressive disorder and emotion state’, *International Journal of Environmental Research and Public Health* **15**(12), 2907.
- Morel, D., Yu, K. C., Liu-Ferrara, A., Suriel, A. J. C., Kurtz, S. G. & Tabak, Y. P. (2020), ‘Predicting hospital readmission in patients with mental or substance use disorders: A machine learning approach’, *International Journal of Medical Informatics* **139**, 1–11.
- Patel, H. & Prajapati, P. (2018), ‘Study and analysis of decision tree based classification algorithms’, *International Journal of Computer Sciences and Engineering* **6**(10), 74–78.
- Paul, M. & Das, S. (2023), ‘Mental health in tech workplace: An analysis’, *International Journal of Science and Research Archive* **10**(01), 221–233.

- Prudenzi, A., Gill, K., MacArthur, M., Hastings, O. & Moukhtarian, T. e. a. (2024), ‘Supporting employers and their employees with mental health conditions to remain engaged and productive at work (mentor): A feasibility randomised controlled trial’, *Journal of Contextual Behavioral Science* **31**, 100720.
- Sammut, C. & Webb, G. (2011), *Encyclopedia of machine learning*, Springer Science Business Media.
- Shafiee, N. S. M. & Mutalib, S. (2020), ‘Prediction of mental health problems among higher education students using machine learning’, *International Journal of Education and Management Engineering (IJEME)* **10**, 1–9.
- Shatte, A., Hutchinson, D. & Teague, S. (2019), ‘Machine learning in mental health: a scoping review of methods and applications’, *Psychological Medicine* **49**, 1426–1448.
- Su, C., Xu, Z., Pathak, J. & Wang, F. (2020), ‘Deep learning in mental health outcome research: a scoping review’, *Translational Psychiatry* **10**, 116.
- Tate, A., McCabe, R., Larsson, H., Lundström, S., Lichtenstein, P. & Kuja-Halkola, R. (2020), ‘Predicting mental health problems in adolescence using machine learning techniques’, *PLoS ONE* **15**(4), e0230389.
- Taunk, K., De, S., Verma, S. & Swetapadma, A. (2019), A brief review of nearest neighbor algorithm for learning and classification, *in* ‘2019 International Conference on Intelligent Computing and Control Systems (ICCS)’, pp. 1255–1260.